

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Winter Semester Examination – Dec. - 2019**

Branch: Electronics and Telecommunication Engineering
Subject: - Introduction to MEMS (BTEXPE506D)
Date: - 20/12/2019

Sem.: - V
Marks: 60
Time: - 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

(Marks)

- Q.1. a) Compare microelectronics and MEMS (06)
- b) State the different types of pressure sensor. Explain anyone in details. (06)
- Q.2. a) What is piezo resistivity? Explain single crystal silicon as piezo resistive material. (06)
- b) What is substrate? Explain active substrate material and its property. (06)
- Q.3. a) State various PVD techniques. Explain any one technique with diagram. (06)
- b) What is photo lithography? Explain procedure in Lithography. (06)
- Q.4. a) Distinguish between wet and dry etching process. (06)
- b) What is bulk micromachining? Explain process in bulk micromachining. (06)
- Q.5. a) What is stress? State and explain different types of stresses. (06)
- b) Explains the bending of beam with types of support. (06)
- Q.6. a) Describe the procedure involved in the Finite Element Methods. (06)
- b) Explain different types finite elements in FEM. (06)